

CS-600: Software Design and Development

3-1 Short Paper: Connecting Domains and Methodologies

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Improving Application Design with Domain Storytelling and Methodology Selection

Writing code is far from being the only thing required to create high-quality software.

Understanding human behavior and business processes is just as important for this goal to be achieved. This paper uses domain storytelling to refine a software idea into a structured workflow. An application suggested here is aimed at providing feedback on clothing outfits and improving user appearance. Based on the domain story, several possible software development methodologies are reviewed to choose the most proper one for the project.

Application Description Analysis

The proposed idea suggests creating a digital mirror capable of analyzing clothes worn by a particular person and suggesting improvements in outfit combination. Although such a concept sounds rather innovative, there is little technical description of this product, and so certain assumptions must be made. To start with, one would assume that such a mirror is equipped with cameras and sensors to take photographs of the client. Also, the system will likely use image recognition technologies to distinguish certain items of clothing and their characteristics.

Artificial intelligence or machine learning techniques will be needed to compare an outfit with certain style guides and trends. In addition, the system is supposed to have profiles of each customer to make recommendations personalized depending on body shape, style, earlier combinations, etc. Finally, it is supposed to access a database of clothes descriptions and style recommendations. Such software would help customers receive immediate feedback and feel confident with their looks. Also, it would serve as help for designers during consultations and

would generally make shopping a better experience for clients. Furthermore, it would speed up decision making and improve customer satisfaction in case of use in a retail environment.

Domain Story: Application Workflow

For better understanding of the described solution, its workflow can be outlined in a brief story format:

1. Client comes up to a digital mirror
2. Mirror takes a picture
3. System detects all clothing items and their colors
4. Artificial intelligence analyzes whether the items fit together
5. Outfit is compared to fashion rules and user preferences stored in the system
6. Feedback is provided to the client in form of recommendations on how to improve appearance
7. User changes outfit according to suggestions of the system
8. Suggestions are at once improved and shown to user
9. Designer consults with customer and recommends further actions

For the above-described process, the following main elements can be distinguished:

- Actors are client, designer, and system
- Work objects are outfit, captured image, and recommendation
- Main activities are picture taking, analysis of obtained data, comparing result with fashion trends, providing feedback, adjusting to suggestions

As seen from the described outline, user interaction is clearly illustrated here.

How Domain Storytelling Helped Describe the Problem Better

Using domain storytelling techniques to describe the problem helped to transform an abstract idea into a much clearer and understandable workflow. Thanks to this technique, it was possible to set up a mutual understanding within the team based on straightforward methods.

As a result, the technique allowed identifying:

- Main actors
- Their actions
- Interactions with work objects

Thus, a clear idea is received of:

- What actions do users perform
- What system does
- Information flow throughout the process

Also, using this technique allowed revealing missing elements needed for the system, such as a possibility to store user information, to use artificial intelligence in operations, and to offer users instant feedback. Another advantage is improving communication between programmers and other specialists since domain storytelling is focused on actual situations rather than complicated diagrams and technical descriptions.

Comparison of Software Development Methodologies

Waterfall

Waterfall methodology suggests sequential completion of one task before moving to the next

one. Such an approach guarantees order but lacks flexibility, making it rather impractical for this project. This is due to requirements related to constantly improving performance of artificial intelligence.

Agile

In contrast to Waterfall, the Agile method is rather flexible and allows constant adjustment of a product based on feedback from testers and users. The iterative process involves short cycles during which certain features are implemented and then tested.

Spiral

Repeating development cycles combined with risk analysis are characteristic of the Spiral method. This approach is quite useful for complex projects requiring the development of advanced artificial intelligence components. Nevertheless, management is usually more complicated compared to Agile.

The recommended Software Development Methodology

Agile method suits this application best of all. The reason is the necessity to improve such aspects of the application as artificial intelligence, interface design, and customization abilities. In addition, Agile provides for developing and implementing a project in stages. For instance, one cycle can be dedicated to recognizing items, and another one to improving the recommendation system. Moreover, the Agile method allows working with stakeholders easily since both users and designers can give feedback at any stage of the process. This is an important advantage since, contrary to Waterfall, this allows changing the project without re-starting it entirely.

Conclusion

Summarizing, it should be said that domain storytelling is an important means to structure software development process. With the help of actors, work objects, and interactions described within a domain story, a clearer understanding of software is reached. Upon comparing different software development methodologies, the Agile method can be considered the most suitable one because of its flexibility.

References

Chandrasekaran, P., & Krishnan, K. (2022). *Domain-driven design with Java: A practitioner's guide*. Packt Publishing.